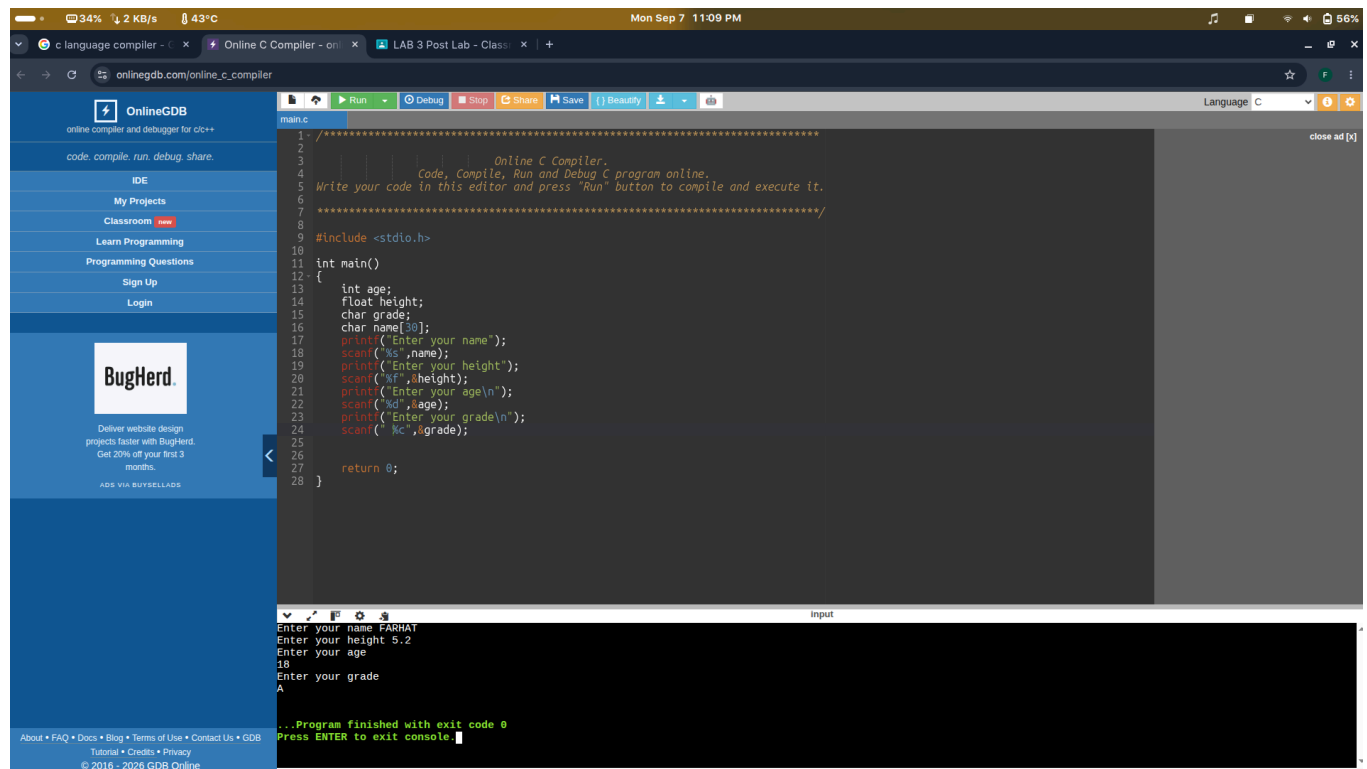


FAST NUCES KARACHI CAMPUS

SE ID (PF)

Task 1 Personal Information:

Write a C program that asks the user to enter their name, age, height, and grade using `scanf()`, and then displays all the entered information using `printf()`. Use the appropriate data types and format specifiers for each type of input.



The screenshot shows the OnlineGDB web interface. The left sidebar contains navigation links: OnlineGDB, code, compile, run, debug, share, IDE, My Projects, Classroom, Learn Programming, Programming Questions, Sign Up, and Login. The main editor area displays a C program for Task 1. The program prompts the user to enter their name, height, age, and grade, and then prints the entered information. The output window at the bottom shows the program's execution with the user's input: FARRAH, 5.2, 18, and A.

```
1- //***** Online C Compiler. *****
2- Code, Compile, Run and Debug C program online.
3- Write your code in this editor and press "Run" button to compile and execute it.
4- *****
5- #include <stdio.h>
6-
7- int main()
8- {
9-     int age;
10-     float height;
11-     char grade;
12-     char name[30];
13-     printf("Enter your name");
14-     scanf("%s", name);
15-     printf("Enter your height");
16-     scanf("%f", &height);
17-     printf("Enter your age\n");
18-     scanf("%d", &age);
19-     printf("Enter your grade\n");
20-     scanf("%c", &grade);
21-
22-     return 0;
23- }
```

Input:

```
Enter your name FARRAH
Enter your height 5.2
Enter your age
18
Enter your grade
A

...Program finished with exit code 0
Press ENTER to exit console.
```

Task 2 Data Types:

Write a C program that creates variables for age using `int`, temperature using `float`, percentage using `double`, and grade using `char`. Take values for all variables from the user using `scanf()` and display the values using `printf()` with the correct format specifiers.

Write a C program that takes the length and width of a rectangle from the user and calculates its area. Display the length, width and area using printf ().

The screenshot shows the OnlineGDB web interface. On the left is a blue sidebar with navigation links: OnlineGDB, code.compile.run.debug.share., IDE, My Projects, Classroom, Learn Programming, Programming Questions, Sign Up, and Login. Below these is a BugHerd advertisement. The main area is a code editor with a dark theme, showing a C program named 'main.c'. The code calculates the area of a rectangle by taking user input for length and width, then printing the calculated area. The program has been executed, and the output is visible in a terminal window at the bottom. The terminal shows the input '4.5' for length and '3' for width, resulting in an area of '13.500000'. The program finished with exit code 0.

```
1- /****** Online C Compiler.
2-                                     Code, Compile, Run and Debug C program online.
3-                                     Write your code in this editor and press "Run" button to compile and execute it.
4-                                     *****/
5-
6-
7-
8-
9- #include <stdio.h>
10-
11- int main()
12- {
13-     float length;
14-     float width;
15-     float area;
16-     printf("Enter length of rectangle");
17-     scanf("%f",&length);
18-     printf("Enter width of rectangle");
19-     scanf("%f",&width);
20-     area=length*width;
21-     printf("length: %f\n",length);
22-     printf("width: %f\n",width);
23-     printf("area: %f\n",area);
24-     return 0;
25- }
```

Input

```
Enter length of rectangle 4.5
Enter width of rectangle 3
length: 4.500000
width: 3.000000
area: 13.500000

...Program finished with exit code 0
Press ENTER to exit console
```

Task 4 Circle:

Write a C program that takes the radius of a circle from the user and calculates its area and circumference. Use float or double for the calculations and display the results using appropriate format specifiers.

The screenshot shows the OnlineGDB web interface. On the left is a sidebar with navigation links: OnlineGDB, code, compile, run, debug, share, IDE, My Projects, Classroom, Learn Programming, Programming Questions, Sign Up, and Login. The main area displays a C program in a text editor. The program calculates the area and circumference of a circle based on a user input radius. The output console at the bottom shows the results for a radius of 4.3.

```
1: /* ***** Online C Compiler. *****  
2: | Code, Compile, Run and Debug C program online.  
3: | Write your code in this editor and press "Run" button to compile and execute it.  
4: | ***** */  
5: #include <stdio.h>  
6:  
7: int main()  
8: {  
9:     float radius;  
10:    float circumference;  
11:    float area;  
12:    printf("Enter the radius of circle");  
13:    scanf("%f",&radius);  
14:    area= 3.142*radius*radius;  
15:    circumference= 2*3.142*radius;  
16:    printf("Area of circle: %f\n",area);  
17:    printf("Circumference of circle: %f\n",circumference);  
18: }  
19:  
20:  
21:  
22:
```

Input:
Enter the radius of circle 4.3
Area of circle: 58.095585
Circumference of circle: 27.021202
...Program finished with exit code 0
Press ENTER to exit console.

Task 5 Shopping Bill:

Write a C program that takes the price and quantity of three different items from the user, calculates the cost of each item and the total bill, and displays the complete bill using printf(). Use appropriate data types because item prices may contain decimal values.

The screenshot shows the OnlineGDB website interface. The left sidebar contains navigation links: OnlineGDB, code, compile, run, debug, share, IDE, My Projects, Classroom, Learn Programming, Programming Questions, Sign Up, and Login. The main area displays a C program in a code editor. The program is a simple calculator for a total bill based on three items. The output window shows the results of a test run with sample input values.

```
1- /*=====
2-
3- Online C Compiler.
4- Code, Compile, Run and Debug C program online.
5- Write your code in this editor and press "Run" button to compile and execute it.
6- =====*/
7-
8- #include <stdio.h>
9-
10-
11- int main()
12- {
13-     int Q1;
14-     int Q2;
15-     int Q3;
16-     float P1;
17-     float P2;
18-     float P3;
19-     float Total_bill;
20-     printf("Enter price and quantity for item 1");
21-     scanf("%f %d", &P1, &Q1);
22-     printf("Enter price and quantity for item 2");
23-     scanf("%f %d", &P2, &Q2);
24-     printf("Enter the price and item for item 3");
25-     scanf("%f %d", &P3, &Q3);
26-     Total_bill = (Q1 * P1) + (Q2 * P2) + (Q3 * P3);
27-     printf("Total_bill: %f\n", Total_bill);
28- }
29- }
```

Input

```
Enter price and quantity for item 1 50.5 2
Enter price and quantity for item 2 60.4
Enter the price and item for item 3 30.7 9
Total_bill: 617.300049

...Program finished with exit code 0
Press ENTER to exit console.
```

Task 06 Find the errors from the following codes (Find the error and rectify)

```
#include <stdio.h>
```

```
int main() {  
    int age;
```

```
    printf("Enter your age: ");  
    scanf("%f", &age);
```

```
    printf("Your age is: %f\n", age);
```

```
    return 0;
```

```
}
```

```
#include <stdio.h>
```

```
int main() {  
    float radius, area;
```

```
    printf("Enter radius: ");  
    scanf("%d", &radius);
```

```
    area = 3.14 * radius * radius;
```

```
    printf("Radius = %d\n", radius);  
    printf("Area = %d\n", area);
```

```
    return 0;
```

```
}
```

The screenshot shows the OnlineGDB website interface. The browser tabs include 'c language compiler', 'Online C Compiler', 'LAB 3 Post Lab - Class', and 'PF 1D Tasks - Google'. The address bar shows 'onlinegdb.com/online_c_compiler'. The left sidebar contains navigation links: 'OnlineGDB', 'code, compile, run, debug, share.', 'IDE', 'My Projects', 'Classroom', 'Learn Programming', 'Programming Questions', 'Sign Up', and 'Login'. The main editor area displays a C program with line numbers 1 to 18. The code is as follows:

```
1 //.....  
2  
3 Online C Compiler.  
4 Code, Compile, Run and Debug C program online.  
5 Write your code in this editor and press "Run" button to compile and execute it.  
6  
7 ...../  
8  
9 #include <stdio.h>  
10  
11  
12 int main()  
13 {  
14     int age;  
15     printf("Enter your age: ");  
16     scanf("%d", &age);  
17     printf("Your age is: %d\n", age);  
18 }
```

Below the code editor is an 'input' section with the text 'Enter your age: 14' and 'Your age is: 14'. At the bottom, a console output shows '...Program finished with exit code 0' and 'Press ENTER to exit console'.

```
1- /***** Online C Compiler.
2- Code, Compile, Run and Debug C program online.
3- Write your code in this editor and press "Run" button to compile and execute it.
4- *****/
5- #include <stdio.h>
6- int main()
7- {
8-     float radius, area;
9-     printf("Enter radius: ");
10-    scanf("%f", &radius);
11-    area = 3.14 * radius * radius;
12-    printf("Radius = %f\n", radius);
13-    printf("Area = %f\n", area);
14-    return 0;
15- }
```

Enter radius: 3
Radius = 3.000000
Area = 28.260000

...Program finished with exit code 0
Press ENTER to exit console.

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Task 07 understand this program and find errors it has many errors (find the errors and rectify them)

```
#include <stdio.h>

int main() {
    char name[30];
    char gender;
    int age;
    int marks;
    float cgpa;

    printf("Enter name: ");
    scanf("%d", name);

    printf("Enter age: ");
    scanf("%f", &age);

    printf("Enter gender: ");
    scanf("%s", &gender);

    printf("Enter marks: ");
    scanf("%f", &marks);

    printf("Enter CGPA: ");
    scanf("%d", &cgpa);

    printf("\n--- Student Information ---\n");
    printf("Name: %d\n", name);
    printf("Age: %f\n", age);
    printf("Gender: %s\n", gender);
    printf("Marks: %f\n", marks);
    printf("CGPA: %d\n", cgpa);

    return 0;
}
```

```
1- /***** Online C Compiler.
2-      Code, Compile, Run and Debug C program online.
3-      Write your code in this editor and press "Run" button to compile and execute it.
4-      *****/
5-
6-      #include <stdio.h>
7-
8-      int main()
9-      {
10-
11-          char name[30];
12-          char gender;
13-          int age;
14-          int marks;
15-          float cgpa;
16-          printf("Enter name: ");
17-          scanf("%s", name);
18-          printf("Enter age: ");
19-          scanf("%d", &age);
20-          printf("Enter gender: ");
21-          scanf("%c", &gender);
22-          printf("Enter marks: ");
23-          scanf("%d", &marks);
24-          printf("Enter cgpa: ");
25-          scanf("%f", &cgpa);
26-          printf("\n--- Student Information ---\n");
27-          printf("Name: %s\n", name);
28-          printf("Age: %d\n", age);
29-          printf("Gender: %c\n", gender);
30-          printf("Marks: %d\n", marks);
31-          printf("CGPA: %f\n", cgpa);
32-          return 0;
33-      }
34-  }
```

Input:

```
Enter name: FARHAT
Enter age: 18
Enter gender: F
Enter marks: 542
Enter cgpa: 3.6

--- Student Information ---
Name: FARHAT
Age: 18
Gender: F
Marks: 542
CGPA: 3.666666
```

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Task 08 Answer the following terms

1. `#include<stdio.h>` why we use this at the top: *It includes the Standard Input Output library, Which allow the program to use basic function.*
2. `main() {}` why we write the code inside this main : *It is the entry point of C program , execution always begin here.*
3. Difference between float & double : *FLOAT stores single precision floating point number(4 bytes of memory) ,whereas DOUBLE stores double precision floating point numbers with higher accuracy(8 bytes of memory).*
4. `scanf` is used for what : *used to take input from user through keyboard.*
5. `printf` is used for what : *used to display output or print text onto the screen .*

Task 09 Answer the following formats

1. %d : Used for integer data types(int)
2. %f : Used for floating-point data types(float)
3. %c : Used for single character(char)
4. %s : Used for string of character(char)
5. %p: Used to print pointer address.

Task 10 explain in one line

- GitHub : A cloud-based platform where developers host,backup and collaborate on Git repositories online.
- Git : A version control system used to track changes in source code during software development.
- Commit :A saved snapshot of your code changes in the repository history.
- Pull request :A request to review and merge code changes from one branch into another.
- Branch :An independent line of development used to wok on new features without affecting the main code.
- Merge : Process of combining changes from different branches into a single branch.